

Motivation

Knowledge distillation (KD)

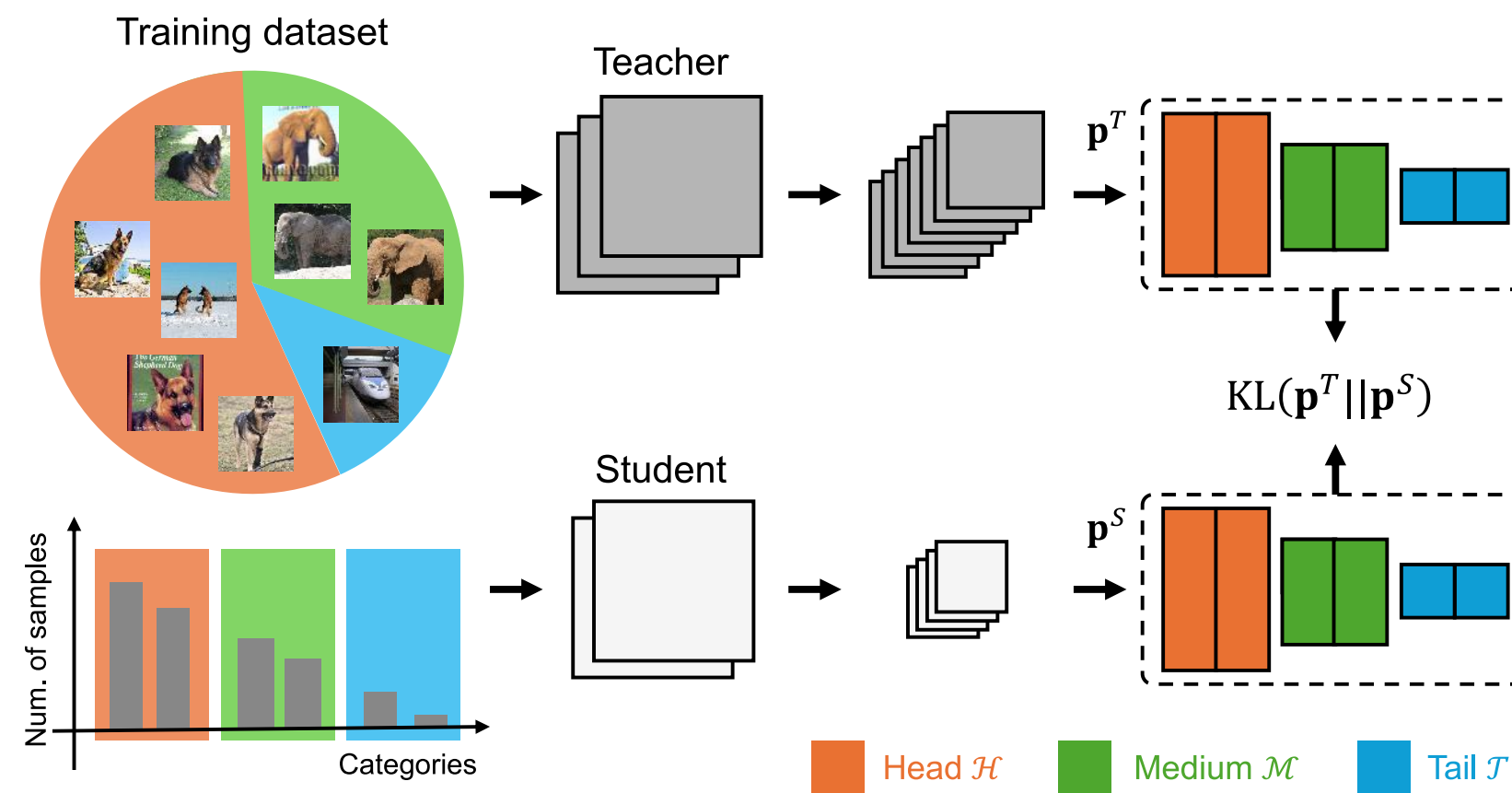
- Model compression: large teacher (T) $\rightarrow$ compact student (S)
- Conventional KD assumes **balanced datasets**.

Can a teacher trained on imbalanced data still offer trustworthy supervision?

Long-tailed dataset

- Common in real-world data
- Head class biased teacher
- Poor performance on tail classes ($\because$ insufficient exposure)

$\rightarrow$ **Failure of standard KD**



Preliminaries

Notation and definition

- For classification with C classes, predictive probability vector, $\mathbf{p} = [p_1, p_2, \dots, p_C] \in \mathbb{R}^C$

$$p_i = \sigma(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)}$$

- Under long-tailed distributions, $C \rightarrow \mathcal{G} \in \{\mathcal{H}, \mathcal{M}, \mathcal{T}\}$

$$\text{KD} = \text{KL}(\mathbf{p}^T \parallel \mathbf{p}^S) = \sum_{\mathcal{G}} \sum_{i \in \mathcal{G}} p_i^T \log \left(\frac{p_i^T}{p_i^S} \right)$$

- Cross-group probability, $\mathbf{p}_{\mathcal{G}} = [p_{\mathcal{H}}, p_{\mathcal{M}}, p_{\mathcal{T}}] \in \mathbb{R}^3$

$$p_{\mathcal{G}} = \frac{\sum_{i \in \mathcal{G}} \exp(z_i)}{\sum_{j=1}^C \exp(z_j)}$$

- Within-group probability, $\tilde{\mathbf{p}}_{\mathcal{G}} = [\tilde{p}_{\mathcal{G}_1}, \tilde{p}_{\mathcal{G}_2}, \dots, \tilde{p}_{\mathcal{G}_i}]_{i \in \mathcal{G}} \in \mathbb{R}^{|\mathcal{G}|}$

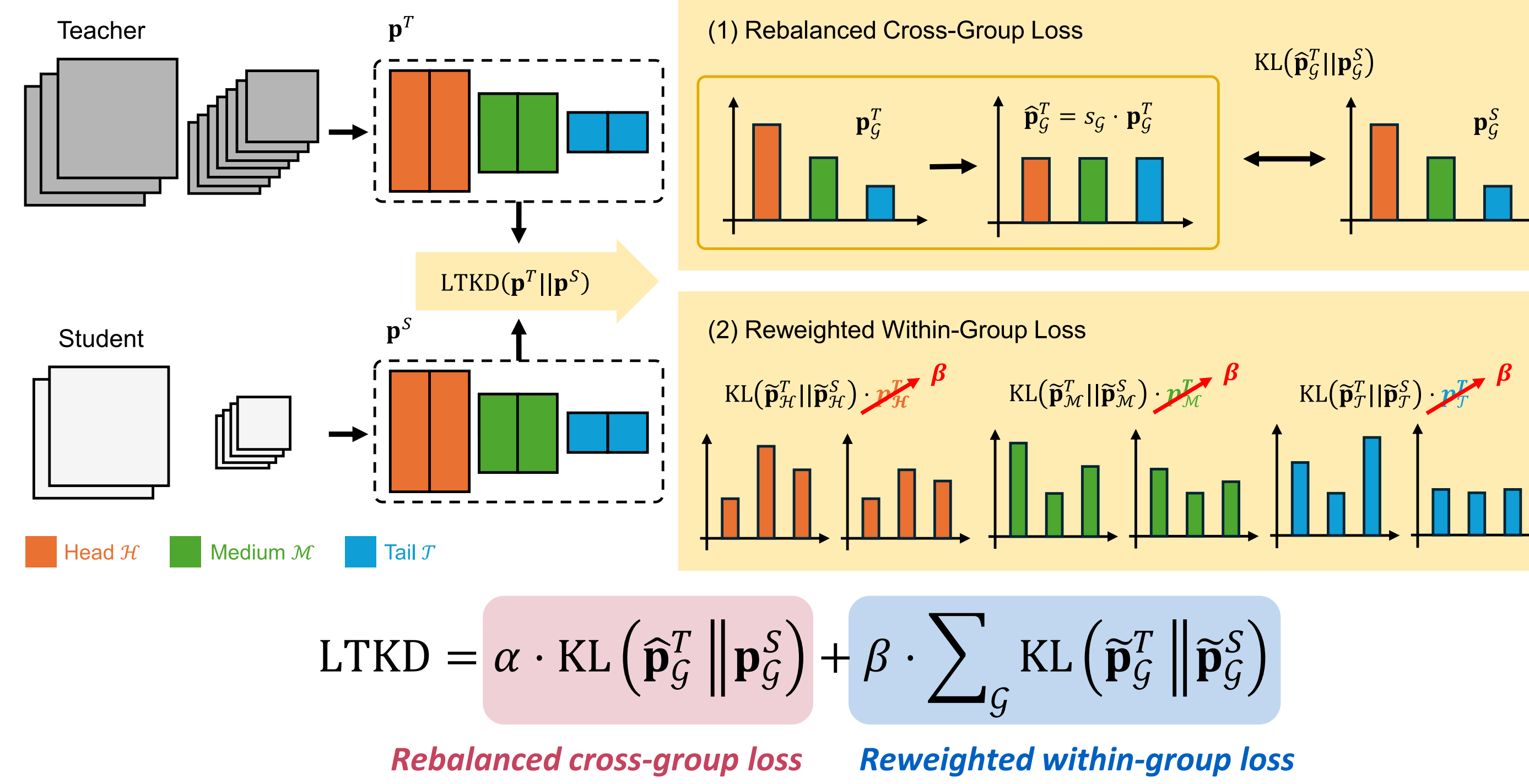
$$\tilde{p}_{\mathcal{G}_i} = \frac{\exp(z_i)}{\sum_{j \in \mathcal{G}} \exp(z_j)}$$

Revisiting KL divergence

$$\begin{aligned} \text{KD} &= \sum_{\mathcal{G}} \sum_{i \in \mathcal{G}} p_i^T \log \left(\frac{p_i^T}{p_i^S} \right) + \sum_{\mathcal{G}} \sum_{i \in \mathcal{G}} p_i^T \log \left(\frac{\tilde{p}_{\mathcal{G}_i}^T}{\tilde{p}_{\mathcal{G}_i}^S} \right) \\ &= \sum_{\mathcal{G}} p_{\mathcal{G}}^T \log \left(\frac{p_{\mathcal{G}}^T}{p_{\mathcal{G}}^S} \right) + \sum_{\mathcal{G}} p_{\mathcal{G}}^T \sum_{i \in \mathcal{G}} \tilde{p}_{\mathcal{G}_i}^T \log \left(\frac{\tilde{p}_{\mathcal{G}_i}^T}{\tilde{p}_{\mathcal{G}_i}^S} \right) \end{aligned}$$

$$\therefore \text{KD} = \text{KL}(\mathbf{p}_{\mathcal{G}}^T \parallel \mathbf{p}_{\mathcal{G}}^S) + \sum_{\mathcal{G}} p_{\mathcal{G}}^T \cdot \text{KL}(\tilde{\mathbf{p}}_{\mathcal{G}}^T \parallel \tilde{\mathbf{p}}_{\mathcal{G}}^S)$$

Long-tailed knowledge distillation (LTKD)



Rebalanced cross-group loss

- Rebalancing before distillation

w/ scaling factors for each group

$$s_{\mathcal{H}} = \frac{p_{\text{avg}}^B}{p_{\mathcal{H}}^B}, s_{\mathcal{M}} = \frac{p_{\text{avg}}^B}{p_{\mathcal{M}}^B}, s_{\mathcal{T}} = \frac{p_{\text{avg}}^B}{p_{\mathcal{T}}^B}$$

$$\mathbf{p}_{\text{batch}} = [p_{\mathcal{H}}^B, p_{\mathcal{M}}^B, p_{\mathcal{T}}^B], p_{\text{avg}}^B = \text{Mean}(p_{\mathcal{H}}^B, p_{\mathcal{M}}^B, p_{\mathcal{T}}^B)$$

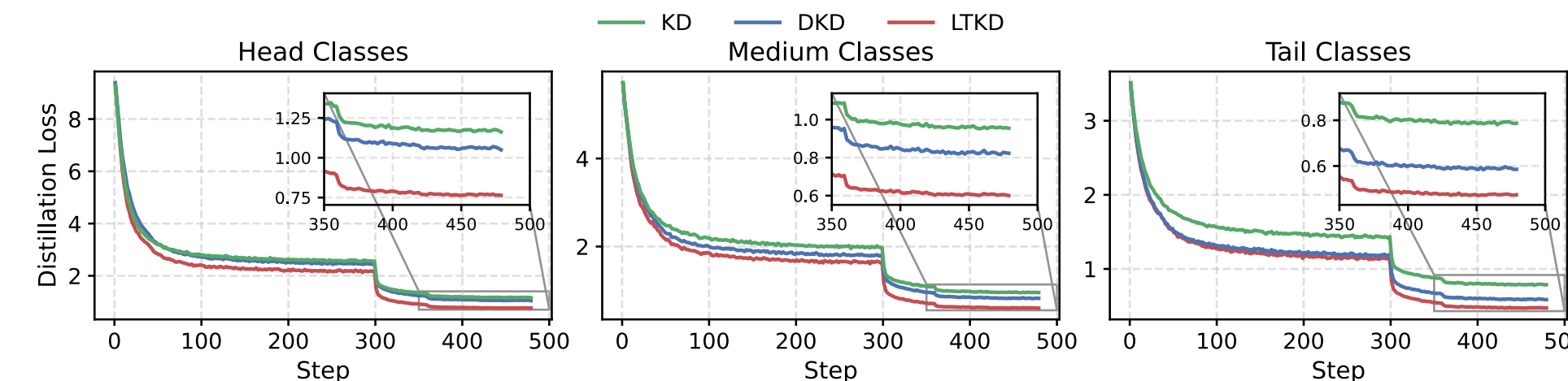
- After **normalization** (for valid probability)

$$\tilde{\mathbf{p}}_{\mathcal{G}}^T = \left[\frac{s_{\mathcal{H}} p_{\mathcal{H}}^T}{\sum_{\mathcal{G}} s_{\mathcal{G}} p_{\mathcal{G}}^T}, \frac{s_{\mathcal{M}} p_{\mathcal{M}}^T}{\sum_{\mathcal{G}} s_{\mathcal{G}} p_{\mathcal{G}}^T}, \frac{s_{\mathcal{T}} p_{\mathcal{T}}^T}{\sum_{\mathcal{G}} s_{\mathcal{G}} p_{\mathcal{G}}^T} \right]$$

Reweighted within-group loss

$$\sum_{\mathcal{G}} p_{\mathcal{G}}^T \cdot \text{KL}(\tilde{\mathbf{p}}_{\mathcal{G}}^T \parallel \tilde{\mathbf{p}}_{\mathcal{G}}^S) = p_{\mathcal{H}}^T \text{KL}(\tilde{\mathbf{p}}_{\mathcal{H}}^T \parallel \tilde{\mathbf{p}}_{\mathcal{H}}^S) + p_{\mathcal{M}}^T \text{KL}(\tilde{\mathbf{p}}_{\mathcal{M}}^T \parallel \tilde{\mathbf{p}}_{\mathcal{M}}^S) + p_{\mathcal{T}}^T \text{KL}(\tilde{\mathbf{p}}_{\mathcal{T}}^T \parallel \tilde{\mathbf{p}}_{\mathcal{T}}^S)$$

- Weaker supervision for tail classes $\rightarrow$ suboptimal convergence



- Equal importance to all groups

by replacing $p_{\mathcal{G}}^T$ with uniform constant β

Main results on ImageNet-LT

- Consistent improvements on the large-scale, imbalanced datasets** as well as general benchmarks such as CIFAR-100-LT and TinyImageNet-LT

T-S Pairs	ResNet34 – ResNet18						ResNet50 – MobileNetV1					
γ	5		10		20		5		10		20	
Group	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All
Teacher	57.75	67.61	50.61	63.85	43.00	59.91	59.62	69.06	52.69	65.57	44.82	61.46
Student	54.48	64.73	47.45	61.18	39.98	57.25	55.42	65.46	49.49	62.55	41.68	58.94
KD [16]	56.14	66.27	49.35	63.03	41.72	59.15	56.58	66.51	50.42	63.70	42.45	59.91
DKD [52]	56.83	66.84	49.95	63.50	42.65	59.82	58.54	68.09	52.39	65.04	45.04	61.46
ReviewKD [6]	<u>57.27</u>	<u>66.96</u>	<u>50.80</u>	<u>63.72</u>	<u>43.48</u>	<u>60.15</u>	<u>58.59</u>	<u>68.10</u>	<u>53.06</u>	<u>65.31</u>	<u>45.35</u>	<u>61.84</u>
DIST [17]	56.79	66.66	50.28	63.56	42.78	59.94	57.29	66.94	50.89	64.09	43.70	60.71
CAT-KD [12]	55.92	66.14	49.58	63.20	42.20	59.67	57.08	66.96	51.00	64.07	43.55	60.54
LTKD	58.33	67.23	52.55	64.29	45.88	60.80	60.17	68.48	54.40	65.52	48.55	62.22
Δ	+1.06	+0.27	+1.75	+0.57	+2.40	+0.65	+1.58	+0.38	+1.34	+0.21	+3.20	+0.38

Ablation study

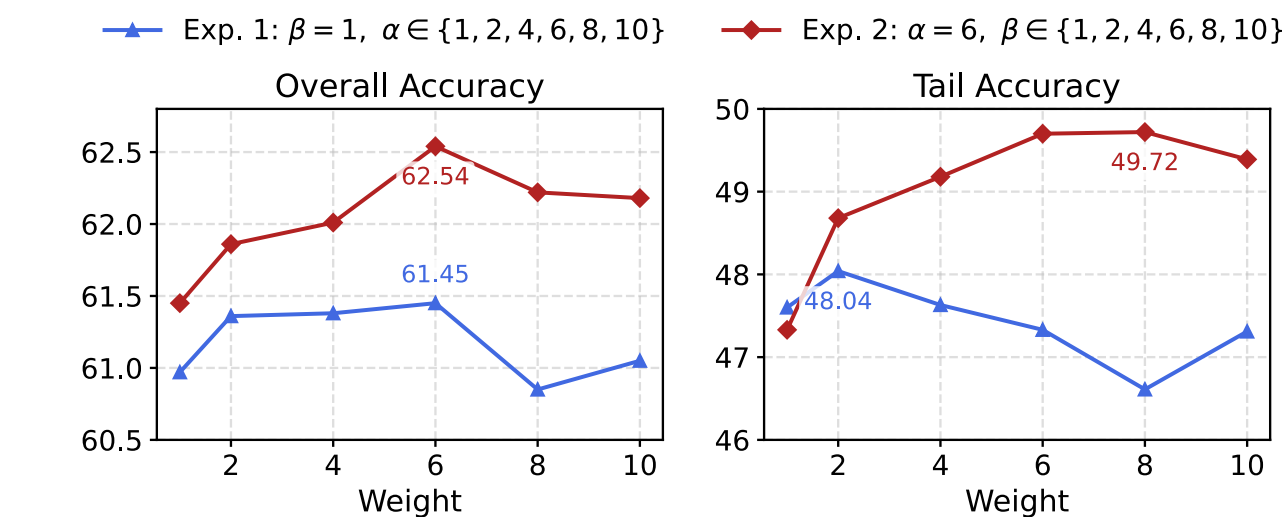
- Cross-group loss:** Non-rebalanced ($\mathbf{p}_{\mathcal{G}}^T$) vs. rebalanced ($\tilde{\mathbf{p}}_{\mathcal{G}}^T$)

Models	Teacher	Student	ResNet32 $\times$ 4				VGG13				WRN-40-2			
			ResNet8 $\times$ 4	ShuffleNetV1	VGG8	MobileNetV2	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1
Loss	Cross	Within	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All
Biased	$\checkmark$	$\times$	38.30	55.77	30.01	48.82	35.55	53.30	23.42	40.29	34.64	53.97	30.23	49.31
Ours	$\checkmark$	$\times$	40.51	56.55	30.68	48.81	37.26	53.70	23.97	40.30	35.23	54.35	32.44	49.91
Δ			+2.21	+0.78	+0.67	-0.01	+1.71	+0.40	+0.55	+0.01	+0.59	+0.38	+2.21	+0.60

- Complementary cross- and within-group components**

Models	Teacher	Student	ResNet32 $\times$ 4				VGG13				WRN-40-2			
			ResNet8 $\times$ 4	ShuffleNetV1	VGG8	MobileNetV2	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1	WRN-40-1	ShuffleNetV1
Loss	Cross	Within	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All	$\mathcal{T}$	All
Baseline	$\times$	$\times$	36.81	57.41	36.13	55.03	37.29	56.13	29.05	47.69	35.63	56.71	38.30	56.61
Ours	$\checkmark$	$\times$	40.51	56.55	30.68	48.81	37.26	53.70	23.97	40.30	35.23	54.35	32.44	49.91
	$\times$	$\checkmark$	42.34	59.78	39.10	56.84	38.54	56.83	31.99	49.20	39.67	58.11	40.45	57.65
	$\checkmark$	$\checkmark$	49.70	62.54	45.94	59.62	45.77	58.86	38.33	52.03	45.74	59.91	48.42	60.94

- Hyperparameters & Number of groups $n(\mathcal{G})$**



Continuous reweighting

$n(\mathcal{G})$	3	4	5	10	20	25	50	100
R32 $\times$ 4–R8 $\times$ 4	51.08	51.08	<u>51.10</u>	51.14	50.99	50.34	50.06	50.41
VGG13–VGG8	47.66	47.85	48.06	48.26	48.69	<u>48.58</u>	48.19	47.82
WRN402–SV1	48.60	48.98	49.03	49.27	49.54	49.90	<u>49.64</u>	47.95
R50–MV2	42.45	43.01	42.99	43.40	43.43	<u>43.42</u>	42.51	41.18